

Notes: Benveniste & Spindt (JFE, 1989)

How Investment Bankers Determine the Offer Price and Allocation of New Issues

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1 Big picture

- **Question.** How do investment bankers use premarket indications of interest to set IPO offer prices and allocate shares, and why does this process generate underpricing?
- **Core mechanism.** Some regular investors have private information about the market value of a new issue. The underwriter designs the offer price and allocation rule so these investors reveal positive information before the IPO price is set.
- **Central friction.** Investors with favorable private information would rather hide it, buy at a lower offer price, and profit in the aftermarket. Truthful revelation requires compensation.
- **Main result.** IPO underpricing is the compensation paid to informed investors for revealing positive information in the premarket.
- **Allocation result.** Investors who indicate strong interest should receive allocation priority. Allocation priority is part of the incentive scheme that makes information revelation credible.
- **Underwriter role.** Underwriters repeatedly interact with regular investors. They can use future allocation priority as a lever to induce current participation and truthful reporting.
- **Contract implication.** Firm-commitment and best-efforts contracts differ because of the underwriter's incentive to presell. Firms with high valuation uncertainty may prefer best-efforts contracts to limit underwriters' preselling incentives.
- **Empirical implications.** Underpricing should rise with the value of investor information, the degree of premarket interest, and the level of presales; regular investors should receive repeated allocation priority.
- **Economic takeaway.** IPO underpricing is not merely a puzzle or pure rent transfer. It is part of an information-acquisition mechanism embedded in the bookbuilding process.

2 Incremental contribution of the paper

1. **Introduces bookbuilding as mechanism design.** Earlier IPO theories emphasized adverse selection, certification, signaling, or lawsuits. This paper models the premarket as an auction/mechanism designed to elicit private information from investors.
2. **Explains underpricing and allocation jointly.** Underpricing is not analyzed separately from rationing. The paper shows that price discounts and share allocations are jointly chosen incentive instruments.
3. **Gives a role to regular investors.** The paper formalizes why underwriters cultivate regular institutional investors: repeated allocation priority creates rents that can be used to obtain truthful information.
4. **Links relationship capital to issuer proceeds.** Underwriter-client relationships with investors are valuable to issuers because they reduce the information rents needed in any one IPO.
5. **Predicts allocation priority.** The model predicts that informed or regular investors should receive favorable allocations, a feature often viewed as favoritism but here interpreted as part of efficient information production.
6. **Connects underwriting contract choice to information risk.** The model gives a theory for why high-uncertainty issuers may use best-efforts contracts and minimum-sales constraints.
7. **Moves IPO underpricing theory toward actual practice.** The paper's mechanism resembles real-world bookbuilding: road shows, indications of interest, conditional pricing, and preferential allocations.

Incremental contribution: The key incremental contribution is to reinterpret IPO underpricing as the cost of eliciting information in a premarket auction. This changes the unit of analysis from the IPO price alone to the full *offer price and allocation* schedule.

3 Data and measurement / model primitives

This is a theoretical paper, so the relevant “data and measurement” section describes model primitives, information states, allocation rules, and contract variables rather than an empirical sample.

3.1 Issuer, issue size, and aftermarket value

A private firm offers a fixed fraction of future cash flows for sale. The issue is divided into Q shares. The full-information aftermarket value depends on the total amount of positive information held by regular investors and the information of occasional investors.

Let H be the number of regular investors. Each regular investor has one signal, good or bad. Let h denote the number of regular investors with good information. A useful reduced-form representation of the conditional expected value is

$$P_h = A - (H - h)\alpha, \quad (1)$$

where α measures the marginal value of one regular investor’s information.

The aftermarket value also includes information of occasional investors:

$$P'_h = P_h + X, \quad (2)$$

where X has mean zero.

Interpretation: The parameter α is central. It is the ex ante marginal value of private information. Larger α means informed investors have more valuable information and therefore require more compensation to reveal it.

3.2 Regular and occasional investors

- **Regular investors** participate repeatedly in IPOs and are targeted in the premarket.
- **Occasional investors** participate in the later market and help set the full-information aftermarket price.
- Each regular investor can buy up to q shares, and the regular investors’ maximum combined demand equals the issue size: $Hq = Q$.
- Each regular has good information with probability p and bad information with probability $1 - p$.

Interpretation: The split between regular and occasional investors captures actual bookbuilding practice: underwriters do not canvass every possible investor before setting the offer price. They rely on a core group of repeated clients.

3.3 Offer price and allocation schedule

The underwriter observes the regular investors’ indications of interest and sets:

- p_h^o : offer price when h investors indicate good information;
- $q_{g,h}$: shares allocated to an investor indicating good information in state h ;
- $q_{b,h}$: shares allocated to an investor indicating bad information in state h .

Shares not sold in the premarket are sold in the aftermarket at the full-information price.

Interpretation: The key decision variable is not just the offer price. It is the complete offer-price-and-allocation rule. This is why allocation priority is economically meaningful.

3.4 Truth-telling and participation constraints

To induce truthful revelation, a regular investor with good information must prefer announcing good information rather than pretending to have bad information. The incentive constraint can be written schematically as

$$E[\text{profit} \mid \text{good signal, announce good}] \geq E[\text{profit} \mid \text{good signal, announce bad}]. \quad (3)$$

Participation also requires nonnegative expected profits from premarket participation:

$$E[\text{premarket profit}] \geq 0. \quad (4)$$

Interpretation: Underpricing exists because the underwriter must satisfy these constraints. If investors could be forced to reveal information costlessly, there would be no need to leave them information rents.

4 Models and theoretical specifications

4.1 Auction-design problem

For a given minimum amount of presales $\bar{Q}$, the underwriter chooses the offer price and allocation schedule to maximize expected proceeds:

$$\max_{p_h^o, q_{g,h}, q_{b,h}} E[\text{issuer proceeds}], \quad (5)$$

subject to truth-telling, participation, and presale constraints.

Interpretation: This is a mechanism-design problem. The underwriter chooses the rule that maps reported interest into prices and allocations.

4.2 Theorem 1: optimal offer-price and allocation schedule

Object	Economic content
Truthful revelation	Investors with good information must receive enough expected profit to reveal it.
Offer price	The offer price must be low enough to reward information revelation.
Allocation priority	Investors indicating good information receive priority allocations.
Underpricing	Underpricing is the cost of paying information rents to informed investors.
Empirical implication	Underpricing rises with the value of private information, premarket interest, and presales.

Notes-created table: Theorem 1 and the optimal premarket mechanism. The original paper contains no tables. This compact table summarizes the mechanism-design logic of Theorem 1.

Economic message: Theorem 1 explains why IPOs are underpriced and why favored investors receive allocations. Both are part of the same incentive scheme.

Interpretation: Underpricing is minimized by allocating priority to investors who reveal good information. Investors with good information are compensated not only through a low offer price but also through a larger allocation when their information is valuable.

4.3 Empirical implications of Theorem 1

The paper highlights four implications:

1. Underpricing is increasing in the ex ante marginal value of investor information.
2. Underpricing is increasing in the level of presales.
3. Underpricing is minimized when priority is given to orders from investors who indicate good information.
4. Underpricing is increasing in the level of interest in the premarket.

Interpretation: The third implication is especially distinctive. Many theories can explain underpricing, but this model also explains preferential allocations as an information-production device.

4.4 Repeated underwriter-investor relationships

The paper extends the model by recognizing that underwriters repeatedly sell IPOs to regular investors. Let L be the present value of future expected profits from remaining in the underwriter's regular-investor network. A regular investor may be willing to accept a loss on a current issue if refusal risks losing future allocation rents:

$$\text{current loss} \leq L. \quad (6)$$

Interpretation: The underwriter's distribution network is an asset. It allows the underwriter to reward investors over time and to punish noncooperation by reducing future allocation priority.

4.5 Theorem 2: underwriter relationships raise proceeds

Mechanism	Implication
Repeated participation	Regular investors expect future IPO rents.
Allocation priority	Future rents can be used as an incentive device in the current IPO.
Threat of exclusion	Investors may buy unattractive current issues to preserve future access.
Issuer benefit	The underwriter can generate higher unconditional proceeds than a one-shot issuer.
Underwriter role	The underwriter is valuable because it pools issues and manages investor relationships.

Notes-created table: Theorem 2 and the value of underwriter relationships. The repeated-game logic gives underwriters leverage that an individual issuer lacks.

Economic message: Theorem 2 gives underwriters a role beyond certification. Underwriters use their repeated relationships with regular investors to lower information rents and improve expected proceeds for issuers.

Interpretation: The theory predicts persistent relationships between underwriters and institutional investors. Regular investors receive rents across deals, but those rents support information revelation and distribution capacity.

4.6 Firm-commitment versus best-efforts underwriting

Under a firm-commitment contract, the underwriter buys unsold shares and therefore has a strong incentive to presell the issue. Preselling reduces placement risk but raises underpricing because more shares must be sold before information is fully reflected in price.

Under a best-efforts contract, the underwriter does not buy unsold shares. This reduces the preselling incentive but increases proceeds uncertainty for the issuer. Minimum-sales constraints can reduce proceeds risk, but they also revive the incentive to presell.

Interpretation: Contract choice balances three forces: valuation uncertainty, issuer risk aversion, and the issuer’s need for guaranteed funds.

5 Identification and theoretical design

This paper is theoretical. Its “identification” is the isolation of a mechanism through assumptions and comparative statics rather than a statistical design.

5.1 What the model isolates

The model isolates the effect of investor private information on IPO pricing and allocation. It deliberately abstracts from other explanations of underpricing, such as signaling, lawsuits, and certification, in order to focus on the bookbuilding mechanism.

Investor information $\rightarrow$ premarket indications $\rightarrow$ offer price and allocation $\rightarrow$ underpricing. (7)

Interpretation: The theory asks: if regular investors possess information that the issuer and underwriter need, how should the IPO mechanism be designed?

5.2 Why the revelation principle matters

The paper treats indications of interest as messages that can be simplified into reports of good or bad information. The revelation principle lets the authors focus on direct mechanisms that induce truthful reporting.

Interpretation: This makes the model tractable while preserving the core economic problem: informed investors must be compensated for revealing favorable information.

5.3 Why allocation is part of price discovery

A purely price-based mechanism cannot fully solve the information problem because investors care about how many shares they receive. If an investor reveals good news but receives no meaningful allocation, the investor gains little from truth-telling. Allocation priority makes information revelation valuable.

Economic message: Share allocation is not merely post-pricing rationing. It is part of the incentive contract.

5.4 Contract-design implications

The model predicts a selection pattern:

- High price uncertainty firms face greater underpricing under firm commitment and are more likely to use best efforts.
- Highly risk-averse issuers prefer firm commitment because it insures proceeds.

- Issuers with strong financing needs prefer firm commitment because it guarantees funds.
- Best-efforts offerings with minimum-sales constraints can have higher underpricing when the constraint forces more preselling.

Interpretation: Contract choice is endogenous. Observed differences between firm-commitment and best-efforts IPOs cannot be interpreted without accounting for issuer selection.

5.5 Main limitations

- The model reduces investor information to a binary good/bad signal.
- It abstracts from how underwriters infer information from actual road-show demand curves.
- The occasional-investor market is assumed to reveal full information in the aftermarket.
- The repeated-game logic is stylized and does not model the full dynamics of underwriter reputation.
- The model provides empirical implications but does not test them directly.

Interpretation: These simplifications make the mechanism clear: information revelation requires rents, and underwriting relationships can reduce those rents.

6 Tables and figures

The original article contains no figures or empirical tables. The notes therefore include notes-created tables that summarize the model results and empirical implications.

6.1 Notes-created Table 1: Main theoretical objects

Object	Role in the model
α	Marginal ex ante value of a regular investor's private information. Higher α raises required underpricing.
h	Number of regular investors with good information. Determines state-contingent offer price and allocation.
p_h^o	Offer price conditional on indications of interest. Adjusted upward when reports are more favorable.
$q_{g,h}$	Allocation to investors reporting good information. Used to reward information revelation.
$q_{b,h}$	Allocation to investors reporting bad information. Kept low to reduce lying incentives.
L	Present value of future rents from being a regular investor. Gives underwriters leverage in repeated relationships.
$\bar{Q}$	Minimum presales. More presales reduce distribution risk but increase underpricing.

Notes-created Table 1: Main theoretical objects. This table replaces original visuals because the paper has no original figures or tables.

6.2 Notes-created Table 2: Empirical predictions

Prediction	Economic logic
Underpricing rises with information value	More valuable private information requires larger information rents to induce revelation.
Underpricing rises with presales	More shares sold before the aftermarket require more compensation to premarket investors.
Allocation priority goes to strong-interest investors	Priority makes truthful revelation of favorable information privately valuable.
Regular investors receive repeated allocations	Future allocation rents give underwriters leverage over current investor behavior.
Best-efforts contracts are more common for high-uncertainty issues	They limit underwriters' incentive to presell the entire issue under firm commitment.
Minimum-sales constraints raise underpricing in best-efforts offerings	They force more preselling and therefore require larger information rents.

Notes-created Table 2: Empirical predictions. The table summarizes the testable implications emphasized in the paper.

7 Short summary

- The paper models IPO bookbuilding as an auction/mechanism for eliciting investor information.
- Regular investors have private information about the aftermarket value of the IPO.
- Investors with good information would prefer to hide it unless compensated.
- Underpricing is the compensation for truthful revelation of favorable information.
- Allocation priority to investors indicating strong interest helps induce truth-telling.
- Underwriters use repeated relationships and future allocation rents to discipline regular investors.
- The model explains why underwriters cultivate regular investors and why IPO allocations are not pro rata.
- Firm-commitment contracts create strong preselling incentives; best-efforts contracts reduce those incentives but increase proceeds risk.
- Minimum-sales constraints in best-efforts offerings can raise underpricing by forcing more presales.
- The main contribution is to connect IPO underpricing, bookbuilding, allocations, and underwriting contract choice in one mechanism.

8 Conclusion

8.1 Core conclusion

Benveniste and Spindt argue that IPO underpricing is a natural consequence of the information-production process in bookbuilding. Informed investors must be compensated for revealing favorable information before the offer price is set. The compensation appears as a combination of underpricing and favorable share allocation.

8.2 Economic intuition

If investors reveal good information, the offer price rises, reducing their potential profit. Without compensation, they would conceal favorable information and buy at a low price. The underwriter solves this incentive

problem by giving truthful investors both expected profits and allocation priority. Thus an IPO's offer price and allocation schedule jointly implement information revelation.

8.3 Role of underwriters

The underwriter is valuable because it repeatedly interacts with investors. A single issuer cannot credibly promise future allocations, but an underwriter can. This future-allocation power lets the underwriter reduce the cost of inducing current investors to participate and reveal information.

8.4 Incremental contribution revisited

The paper shifts IPO theory from the question "Why are IPOs underpriced?" to the richer question "How does the bookbuilding mechanism use price and allocation to elicit information?" This distinction is important because the model explains stylized facts about both prices and allocations.

8.5 Implications

The theory implies that preferential allocations to regular investors may be part of an efficient mechanism rather than pure favoritism. It also implies that underpricing should be related to premarket demand and valuation uncertainty, and that the choice between firm commitment and best efforts is shaped by the underwriter's preselling incentive.

8.6 Limitations

The paper is stylized. It abstracts from multi-dimensional information, bookbuilding conversations, legal constraints, aftermarket stabilization, and the full strategic relationship among issuers, underwriters, and institutions. It also provides predictions rather than direct empirical tests.

8.7 Takeaway

The enduring lesson is that IPO underpricing can be the price of information. Bookbuilding is not merely a way to sell shares; it is a mechanism for purchasing and aggregating investor information before the firm goes public.